



Challenges, paradoxes and lessons for inclusive AI governance: an interdisciplinary analysis of corporate initiatives

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Abstract

Current AI governance strategies and guardrails ensure that AI systems adhere to the ethical principles, legal requirements, and safety standards. However, they create organisational rigidity which could potentially delay new deployments, fail to capture tacit qualitative information, are sometimes unable to enforce their policies, and lack binding corporate commitments. This research aims to address these weaknesses by employing a multi-case analysis involving a sector-level critical synthesis to examine corporate governance and social responsibility (CG/SR) initiatives across the oil and pharmaceutical industries to provide interdisciplinary understanding for the ongoing global dialogue on inclusive and sustainable AI governance. The findings uncover practices, challenges, and paradoxes in corporate governance initiatives across the two industries and their implications. Drawing actionable insights from the findings, we propose four (4) elements that could be used to enhance/reframe existing AI governance frameworks across national, regional/multinational and organisational contexts. We also offer practical pathways for putting our innovative AI governance framework to work in the real-world, which could turn AI principles into operational strategies for AI/technology governance.

Keywords Responsible AI · AI governance · Corporate governance · AI strategies · Case study · Critical synthesis · Technology governance

1 Introduction

In August 2025, the United Nations General Assembly established the Global Dialogue on AI Governance to ensure that global “governance reflects the priorities of all nations,

not just the most technologically advanced and that the benefits of AI are shared by all” [1]. This initiative, which was one of the commitments adopted at the 2024 Summit of the Future at the United Nations, is organised on the wheels of the increasing global debates on AI Ethics and Governance practices across corporations and institutions, and their consequential socioeconomic opportunities and risks to the daily lives of the people, societies and environments [2, 3]. These conversations and advocacies which traverse national and international boundaries emphasize the need to adopt acceptable and appropriate ethical regulatory practices – accountability, trustworthiness, fairness, inclusion, human oversight and sustainability to the design and deployment of AI technologies [4–6]. This has been demonstrated by the avalanche of AI governance strategies across national, regional/multinational and organisational contexts [7, 8] with specific focus on risk-based governance, regulatory principles and a mixture of risk-based and regulatory principles of AI, respectively (See Table 1).

Despite growing interest, current AI governance strategies and guardrails have drawbacks. They create organisational rigidity which could potentially delay new

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Table 1 Summary of AI governance strategies (frameworks) drawn from major national, regional/international and corporate institutions

Categories of AI governance strategies	Key components (that facilitate good AI governance practices)	Proprietors of the AI governance frameworks
National AI governance strategies	Risk management, accountability, transparency, human oversight, fairness and non-discrimination, privacy and data governance, safety and security, regulatory compliance, and impact assessment	Examples include the UK AI governance approach, the EU AI Act, the US NIST AI Risk Management Framework (AI RMF), Singapore's Model AI Governance Framework, Nigeria AI Strategy (NAIS), and Canada's Directive on Automated Decision-Making, etc
International AI governance strategies	Human-centered values, transparency and explainability, accountability, fairness and inclusion, safety, security and robustness, stakeholder participation, sustainability and social wellbeing, continuous monitoring, and international cooperation	Examples include the OECD AI Principles, UNESCO Recommendation on the Ethics of AI, ISO/IEC 42001, IEEE Ethically Aligned Design, and the G7 Hiroshima AI Process
Corporate/institutional/organisational AI governance strategies	AI policy, governance structure, AI inventory, risk and impact assessment, data governance, human oversight, model validation, documentation, incident management and continuous improvement	Examples include ISO/IEC 42001 AI Management Systems, Microsoft Responsible AI Standard, Google AI Principles, IBM AI Ethics Board, and enterprise AI governance programmes

deployments, fail to capture tacit qualitative information, are sometimes unable to enforce their policies, and lack binding corporate commitments. Moreover, significant implementation gaps remain due to weak institutional mechanisms and inadequate consideration of social, political, and environmental contexts across the AI value chain. These shortcomings undermine public trust in AI systems and highlight the need for more inclusive governance. Studies show that corporate governance initiatives across the oil and pharmaceutical industries are characterised by processes that keep organisational management accountable, protect stakeholder interests while supporting ethical, legal, and safety objectives [8, 9]. We believe that AI ethics and governance can draw valuable lessons from corporate governance and social responsibility practices in the oil and pharmaceutical industries, where transparency, accountability, legal compliance, stakeholder engagement, and management oversight [8] strengthen organisational trust. Critically examining these established frameworks can inform the development of more inclusive, context sensitive, and effective AI governance strategies, where ethics and governance practices in the AI development and deployment benefit all communities equitably. Such an inclusive AI governance framework would span multiple dimensions, including shareholder participation, demography and equity, cultural, socioeconomic and geographical, technical and operational, as well as legal and human rights dimensions - to ensure wider organisational alignment.

Therefore, this paper explores the corporate governance and social responsibility initiatives within the oil and pharmaceutical industries to uncover subsisting practices, challenges, paradoxes and draw actionable insights that will inform future directions of AI governance strategies. These two industries were selected because they are deeply related through shared high-stakes risk controls and a long history of established corporate governance practices that has

witnessed intense public scrutiny over the years and across geographical contexts. The main research question guiding this analysis is: *What lessons from corporate governance and social responsibility initiatives in oil and pharmaceutical industries can inform more inclusive AI governance strategies?* In this paper, we approach “responsible AI” (RAI) as a salient concept that denotes the “combination of technical solutions and human efforts” towards developing and adopting a set of principles, practices and guidance in ways that are ongoing and evolving [10 p.2180]. We employed a multi-case analysis [11] which involves a sector-level critical synthesis of corporate governance initiatives [12], with the oil and pharmaceutical industries serving as source domains for our conceptual framework. The critical synthesis is meant to determine relevant themes from evidence provided [13]. This method includes a four-phase process: design phase (to define the scope of the study), a withincase analysis phase (a critical sector-level case synthesis of literature from each of the two industries), the crosscase analysis phase (to identify common themes and differences), and followed by interpretation phase (conceptual revision). Literature sources were accessed through handsearching techniques [14, 15] across databases, including Google Scholar, ACM Digital Library, PubMed, OECD AI Policy resources, international reports, and corporate websites of selected oil companies. Adopting and structuring our sector-level critical synthesis of the industries using six (6) components from an established framework that governs corporate initiatives [8], we identified key corporate initiatives across the two industries, alongside challenges and paradoxes, with direct implications for responsible AI. We also proposed four elements that could be used to enhance/reframe existing AI governance frameworks across national, regional/multinational and organisational contexts. As AI and major technology companies face similar structural tensions [16–18], these interdisciplinary findings offer practical insights for

developing responsible, equitable, transparent, secure, safe, and sustainable AI and technology governance.

This article is organised as follows. Section 2 reviews the literature, Sect. 3 outlines the methodology, Sect. 4 presents the case analysis, Sect. 5 discusses AI governance implications and stakeholder roles, and Sect. 6 concludes with the study's limitations, practical implications, and directions for future research.

2 Background literature

In this section, we synthesize relevant literature across our study chain.

2.1 Overview of AI governance frameworks

Artificial Intelligence is a sociotechnical system. The field of social computing emphasizes the need for deeper reflection on the societal implications of digital technologies including AI [19]. From the international institutions [1, 20] to the different National AI Strategies tracked by the OECD.AI Policy Navigator [21], the overarching goal is to develop AI frameworks that guide ethical design and sustainable deployment of AI systems for public good [22]. The assumption is that these frameworks would help to mitigate potential risks and harms that unregulated AI products and services could have on human rights, user privacy and security, the environment and wider society [23–25].

For example, the OECD AI Policy Navigator presents comparative data on AI policies, governance initiatives, and responsible AI developments across the countries of the world [26]. Also, studies show how corporations (IBM, Microsoft, Google, Cisco, Intel, PWC, etc.) and other public institutions (UNESCO, US NIST, UK Alan Turing Institute etc.) have developed, under the broad umbrella of AI4Good, AI governance frameworks and guidelines to mitigate/minimise potential risks associated with the deployment of AI across critical sectors of the global economy [27–30]. The 2023 International Organization for Standardization ISO/IEC 42,001 provides concerted requirements for establishing, implementing, maintaining, and continually improving AI management systems (AIMS) [31]. The United States NIST AI RMF (Artificial Intelligence Risk Management Framework) provides a structured guideline for mapping, measuring, and managing AI-related risks and consequences [32]. The UK's National AI Strategy outlines a ten-year vision for strengthening the UK's position as a global leader in responsible and trustworthy innovation in artificial intelligence [33]. Taken all together, the above efforts have demonstrated the diversity of approaches towards governance and regulation of AI; however, there is the prevalence

of implementation gaps. Below, we summarize the different categories of AI frameworks/strategies, their key components and proprietors of the frameworks.

Table 1 highlights the key components of existing AI governance frameworks. Specifically, national AI governance strategies prioritise risk-based governance, transparency, accountability, and human oversight as core controls, while international strategies place greater emphasis on human rights, inclusiveness, societal wellbeing, and regulatory principles. Corporate AI governance frameworks are generally more comprehensive, integrating both risk-based approaches and regulatory requirements. However, significant gaps remain regarding the effectiveness of centralised versus distributed governance in achieving globally consistent standards across existing legal and technical infrastructures. Experience from industry nationalisation in the Global South demonstrates that weak institutional capacity often undermines policy implementation. Similarly, the current AI resurgence raises questions about whether political commitments will translate into enduring governance mechanisms, including standards, registers, and regulatory protocols.

2.2 Reviews of existing AI governance studies

As highlighted in the preceding section, stakeholders including the UN, UNESCO, OECD, and IEEE have advocated institutional mechanisms for governing and regulating AI research and development [34]. This has prompted proposals for a centralised International Artificial Intelligence Organisation or coordination mechanisms like the Global Partnership on AI [35]. However, AI governance must address global disparities, particularly Africa's limited share of AI computing capacity (1%) and talent (3%). A key question remains: to what extent can centralised or fragmented governance frameworks, such as global AI dialogues and readiness indices, bridge the global AI governance divide [35–37].

First, AI governance, as an emerging regime with periods of rapid advancement and stagnation, operates within a fragmented landscape. Due to AI's black-box challenges, research and policy efforts have largely focused on safety, control, misuse prevention, and compliance. However, the question of whether AI can be "cohesively regulated" given its diverse technologies, applications, and risk profiles continues to generate debate [35, 126]. As a polycentric and fragile system, global AI governance initiatives often reflect political commitments or media narratives rather than institutional capacities with effective monitoring mechanisms [39]. While the race to AI regulation accelerates [40], scholars advocate for anticipatory, context-specific governance models grounded in indigenous values, accountability, and

regional coordination [39, 41]. Social imaginaries and political economy perspectives further challenge purely developmental views of technology [34, 36].

Second, the AI governance divide, like climate governance, risks creating paternalistic relationships where less economically developed nations become dependent on imported interventions. This dependency positions local startups as beta-testers and consumers of foreign-owned infrastructure, as seen in initiatives such as Nigeria's AWARRI LLM, South Africa's Cassava AI Factory, Anthropic's Rwanda AI for Public Good project, and Kenya's ixAfrica cloud [39, 42]. The privatisation of digital infrastructure, including subsea cables, may further enable institutional and regulatory capture, reinforcing technological hegemony [43]. Reliance on proprietary infrastructures raises concerns about accountability, regulatory fragmentation, and extractive innovation models [44]. Across Africa, governance approaches vary, highlighting persistent policy-implementation gaps caused by limited sustainable infrastructure and operational capacity. These challenges require moving beyond viewing technology solely as an instrument for development and towards locally grounded, equitable governance models.

From the above, AI research and innovation have shifted from technical domains to socio-technical and geopolitical dimensions [45]. As AI becomes entangled with technological science and political economy, human-centred AI studies highlight that neglecting social, political, and environmental contexts risks reproducing bias and inequity. Corporate narratives may obscure transparency and accountability, particularly for marginalised communities [46, 47]. These ethical challenges fuel distrust [9], reinforcing calls for more inclusive AI governance and socially responsible approaches that protect diverse communities while recognising their values and needs.

2.3 Corporate governance and social responsibility in the oil and pharmaceutical industries

AI ethics refers to the principles, legal requirements, and safety standards that guide the responsible development and use of AI, while AI governance provides the operational frameworks, processes, and safeguards to ensure compliance. Similar to corporate governance and social responsibility (CG/SR) practices in the oil and pharmaceutical industries, AI governance promotes accountability, transparency, stakeholder trust, and responsible decision-making. Corporate governance ensures that organisations and their leaders remain accountable by establishing appropriate controls and processes that protect stakeholder interests while supporting ethical, legal, and safety objectives [8, 9].

For example, Robinson [48] identified the Open Government Partnership (OGP) and Extractive Industries Transparency Initiative (EITI) as leading global multi-stakeholder initiatives promoting transparency and open governance in the oil and energy sector. With over 30 shared member countries, both frameworks emphasise digital transformation, data-driven policymaking, knowledge sharing, citizen engagement, and sustainable development, aligning closely with AI governance objectives. Similarly, Emeka-Okoli et al. [49] demonstrated that corporate governance frameworks shape corporate social responsibility (CSR) strategies in the oil and gas industry, highlighting the importance of regulatory alignment, stakeholder engagement, and board oversight in achieving sustainability. In the pharmaceutical sector, Wong et al. [50] found that CSR and Environmental, Social, and Governance (ESG) strategies, including equitable pricing, donations, and environmental initiatives, vary according to partner contexts and national capacities. Their findings reinforce the value of public-private collaboration in developing inclusive and sustainable CSR approaches.

However, deceptive practices persist in the implementation of corporate governance and social responsibility initiatives within the oil/energy and pharmaceutical sectors. Using the system generalized method of moments, Silva et al. [51] examined 140 global pharmaceutical companies listed between 2016 and 2022 to identify ESG washing determinants. Their findings reveal that factors such as larger boards, female representation, and ISO 14,001 certification may promote ESG-washing, while experienced directors and greater research and development investments help mitigate such practices.

The findings from CG/SR initiatives in the oil and pharmaceutical industries have significant implications for individuals, societies, and operational communities, contributing valuable insights to ongoing discussions on responsible AI research, ethics, and governance. Given that AI and Big Tech companies face similar structural tensions arising from diverse application domains [16], these lessons offer critical interdisciplinary guidance for developing governance frameworks that are responsible, equitable, transparent, secure, and sustainable. This article proposes reframing AI governance as a dynamic, stakeholder-driven process requiring continuous coadaptation, reflexivity, adaptability, and inclusivity throughout the AI lifecycle.

3 Research methodology

While AI ethics and governance research have been gaining traction in recent years, there is a dearth of interdisciplinary studies which could offer actionable insights to addressing varied tensions associated with the implementation

of responsible AI governance strategies in the real-world. Therefore, this study wants to find out *what sort of learnings could be drawn from the corporate governance and social responsibility initiatives of organisations to inform an enhanced paradigm for inclusive AI governance strategies?* This section presents our research methodology, showing how we selected our literature and a multi-case analysis which involves sector level critical synthesis of corporate governance initiatives, with the oil and pharmaceutical industries serving as source domains for our conceptual framework.

3.1 Literature selection and analysis

Between May 2024 and June 2026, the authors employed the handsearching method to identify over one hundred and twenty (120) relevant sources for this study. Handsearching is a manual approach to literature retrieval that involves systematically reviewing journal issues, conference proceedings, and grey literature sources that may not be consistently indexed in electronic databases. Researchers use handsearching techniques to manually review materials page by page to locate relevant studies that may be missed by electronic database searches (Cochrane Information Specialists' Handbook Section – 5) [14, 15]. This approach was applied across databases, including Google Scholar, ACM Digital Library, OECD AI Policy resources, international reports, and corporate websites of selected oil companies. Unlike electronic database searches, which often return large volumes of results based on keyword indexing, handsearching allows for a more thorough and deliberate inspection of relevant journal articles, conference papers, and abstracts, including those that may lack consistent metadata or are not yet indexed in major databases. Additionally, handsearching complements database searches by reducing retrieval bias and identifying relevant studies overlooked due to metadata inconsistencies, indexing delays, or terminology variations [52, 53].

To guide this process, a range of keywords and thematic combinations were employed, including: “corporate governance+oil industry”, “corporate governance+pharmaceutical industry”, “AI governance”, “AI strategies”, “responsible AI principles”, and related variations. As mentioned, boolean operators and keyword combinations were adapted to each platform where applicable. Inclusion criteria were defined as follows:

1. Publications in English.
2. Focus on corporate governance in the oil and pharmaceutical industries, AI ethics and governance, as well as national, international/regional, and organisational AI policies and strategies.

3. Conceptual and applied research with relevance to inclusive AI governance, legal regulation and compliance.

Grey literature on ethical frameworks and institutional reports of organisations such as OECD AI Policy resources, and corporate websites of selected oil companies (such as Shell Global¹, Petrobras², BP³, TotalEnergies⁴, ExxonMobile⁵, etc.) and pharmaceutical industries were included where they offered conceptual contributions relevant to AI governance in the real-world contexts. In addition to primary sources, review articles were also examined to identify foundational frameworks, methodological trends, and synthesised insights relevant to the intersection of AI governance and corporate governance initiatives. The subsequent sections present a sector-level critical synthesis of corporate governance initiatives, with the oil and pharmaceutical industries serving as source domains for our conceptual framework, which helped to answer our research questions and inform the next sections of this article.

Thus, we employed a multi-case analysis which involves a sector-level critical synthesis of corporate governance initiatives, with the oil and pharmaceutical industries serving as source domains for our conceptual framework (see Fig. 1). These two industries were selected because they are deeply related through shared high-stakes risk controls and a long history of established corporate governance practices that has witnessed intense public scrutiny over the years and across geographical contexts. Specifically, both industries navigate strict environmental, safety, and health mandates, manage massive capital investments with decade-long horizons, focus on environmental, social and governance (ESG), and use specialized board committees to manage catastrophic risks. We believe that the findings from our critical analysis of the corporate governance initiatives within these two industries will offer insights to producing contextually relevant AI governance strategies that are inclusive.

As shown in Fig. 1, the multi-case analysis which involves a sector-level critical synthesis of corporate governance initiatives in the two industries is a four-phase process: In phase 1 (design phase), the research scope, theoretical framework, and case subjects are defined based on the research questions. In phase 2 (within-case analysis), a sector-level critical synthesis of literature is conducted and

¹ <https://www.shell.com/sustainability/our-approach/commitments-policies-and-standards/business-integrity.html>.

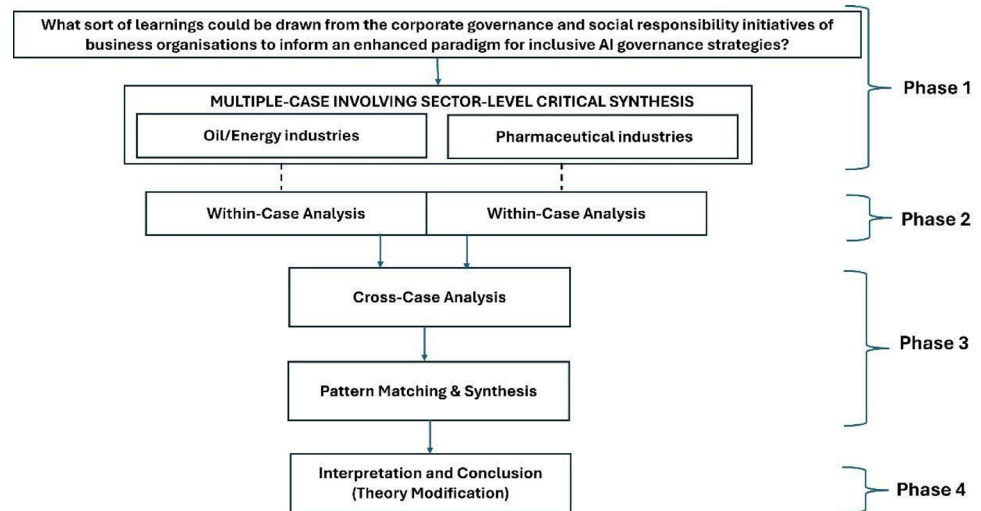
² <https://petrobras.com.br/en/sustentabilidade/governanca-corporativa>.

³ <https://www.bp.com/en/global/corporate/who-we-are/our-beliefs-and-code-of-conduct.html>.

⁴ <https://totalenergies.com/code-conduct-0>.

⁵ <https://corporate.exxonmobil.com/who-we-are/our-global-organization/corporate-governance/code-of-ethics>.

Fig. 1 A multi-case analysis involving- sector-level critical synthesis diagram for oil and pharmaceutical industries



descriptive reports are produced. In phase 3 (cross-case analysis), we compare the corporate governance initiatives in both industries to identify similarities, mechanisms, processes, and governance practices. In phase 4 (interpretation phase), the findings from the sector level critical synthesis are used to refine the initial framework, leading to conceptual generalisations and recommendations. While this approach highlights operational challenges and conceptual complexities within both industries, it also offers useful insights to the real-world implementation of AI ethics and governance principles and policies.

4 Within-case analysis: evaluation of the oil & pharmaceutical industries

This section presents concise yet comprehensive assessments of corporate governance and social responsibility initiatives within the oil and pharmaceutical industries. We identify recurring themes and examine how these practices are implemented across organisational contexts.

4.1 The oil/energy industry—a critical case synthesis

Unlike the emerging responsible AI industry, the global petroleum industry has undergone major transformations over the past two centuries. Following the commercial drilling of the Drake Well in Pennsylvania, USA, in 1859 [54–56] and subsequent oil discoveries during the 1950s and 1960s [57], increasing concerns over transparency, accountability, and sustainability prompted the introduction of regulatory policies and industry reforms [58]. A major milestone was the establishment of the Organization of Petroleum Exporting Countries (OPEC) in 1960, which harmonized petroleum policies among member states, strengthened

governance, stabilized crude oil production, and influenced global petroleum pricing [58, 59].

Studies show that nationalisation has driven the development of national and international policies, legal frameworks, and regulations to promote ethical practices in the oil industry. In Nigeria, policy reforms have focused on improving transparency, attracting foreign investment, strengthening local content, and addressing environmental degradation in the Niger Delta, where oil exploration has severely affected community livelihoods [60]. These reforms led to the establishment of the Niger Delta Development Commission (NDDC) in 2000 to address the region's socio-economic and environmental challenges [57]. Similarly, other OPEC member states have adopted regulatory frameworks that promote ethical practices, stakeholder engagement, and sustainable industry–community relations [57, 58, 61].

Environmental responsibility in the oil industry has been widely studied due to its climate and ecological impacts. Research has examined the environmental effects of oil drilling, refining, and transportation on ecosystems, water quality, and air pollution [61, 63, 64]. Akpan [65] evaluated oil spill response strategies, control measures, and cleanup technologies [65]. Other studies highlighted how national and international regulations, including strict environmental policies [66, 67] and agreements such as the Paris Agreement [68], strengthen prevention, mitigation, and environmental responsibility.

Furthermore, studies have examined how the Oil industry integrates corporate social responsibility (CSR) into its operations, exploring the motivations, impacts, and challenges of balancing environmental responsibility with financial performance [69, 70]. These studies assessed initiatives including energy efficiency, waste management, biodiversity conservation, community engagement, and the adoption of renewable energy and alternative fuels to mitigate

environmental impacts [69–71]. They also explored the ethical principles guiding corporate environmental practices. The Russia–Ukraine war has further accelerated Europe’s search for alternative energy sources, renewing interest in the Trans-Saharan Gas Pipeline⁶. However, the project continues to face logistical and security challenges rooted in inadequate stakeholder inclusion, responsible governance, and sustainable engagement with marginalized communities affected by oil exploration.

4.2 The pharmaceutical industry—a critical case synthesis

The pharmaceutical industry demonstrates how scientific innovation, collaboration, and long-term commitment can improve global health. It has established robust standards for drug safety and efficacy through strict regulatory compliance, rigorous testing and validation, continuous post-market monitoring, and ongoing advances in research and technology [72–75]. Increasingly, pharmaceutical companies are balancing commercial objectives with investments in areas that generate limited financial returns but address critical unmet medical needs. This approach is particularly evident in public–private partnerships, where governments, non-profit organisations, academic institutions, and industry collaborate to accelerate research and development with significant public health benefits [76]. The World Health Organization also highlights the industry’s use of incentive mechanisms, including tax credits, grants, and subsidies, to encourage investment in research for neglected diseases and conditions disproportionately affecting underserved populations, demonstrating how corporate innovation can advance both business objectives and global health equity [77].

A review of pharmaceutical operations, particularly in low- and middle-income countries, reveals persistent transparency and accountability challenges, including monopolistic practices, systemic bribery, and conflicts of interest [78]. Corruption involving drug registration, procurement processes, and facility inspections has been widely reported [79, 80]. The pharmaceutical industry’s role in perpetuating these challenges cannot be overlooked, as unethical practices further strain already fragile healthcare systems in Africa. Collusion between pharmaceutical representatives and public or private healthcare institutions has contributed to prolonged drug price inflation, limiting the affordability and accessibility of essential medicines [81]. However, stakeholders are increasingly working with governments and organisations to improve pharmaceutical affordability by strengthening price monitoring mechanisms across medicine supply chains and ensuring compliance with agreed

pricing frameworks between pharmaceutical companies and procurement agencies.

From an ethical and environmental stewardship perspective, the pharmaceutical industry increasingly prioritises patient privacy and data security due to the sensitive nature of medical research and innovation. However, Big Pharma continues to face criticism over data bias and underrepresentation in clinical trials [82, 83], contributing to unequal healthcare outcomes. Vulnerable populations in many African communities have historically experienced some of the most unethical clinical trials recorded, raising serious concerns about research integrity, safety, and respect for human rights [84, 85]. Notable examples include Pfizer’s Trovan (trovafloxacin) trial for bacterial meningitis in 1996 [86] and early HIV antiretroviral trials in Africa [87], which highlight the consequences of inadequate ethical oversight. Such practices can reinforce healthcare disparities and undermine public trust. Although ethical compliance is often challenging and resource-intensive, it remains essential to ensure transparent consent and independent oversight, rather than relying on vulnerable approval systems susceptible to undue influence [88]. Pharmaceutical companies must uphold global data protection standards, safeguard patient information, and protect intellectual property while ensuring that clinical research prioritises human dignity and safety.

The pharmaceutical industry is increasingly adopting green chemistry and sustainable manufacturing to reduce the environmental impact of drug production [89]. Companies are designing safer synthetic routes, minimising hazardous chemicals, optimising energy-efficient reactions, and implementing waste management strategies to enhance recycling and reuse. To support Net-Zero healthcare, leading firms are also investing in renewable energy technologies, including solar, wind, and biomass systems, to generate cleaner onsite energy and reduce carbon emissions [89–91].

Although there are reported ethical and societal issues across the pharmaceutical industry, this sector boasts of some exemplary practices worth emulating by other sectors [92, 93]. It is a practical example of translating outcomes of intense research into beneficial therapeutic interventions. Adopting a strategy of continuous interest in the lifecycle of medical services, pharmaceutical companies continuously monitor the safety and efficacy of their inventions, even after they have been approved and marketed. This involves collecting and analysing real-world data, conducting post-marketing studies, and updating product labels as needed based on new information [94]. The pursuit of rigorous science and cutting-edge technology to inform new interventions, unparalleled compliance with regulation and actively partnering with governments and stakeholders to widen access to medicines [95] stands the pharmaceutical industry out.

⁶ <https://www.aljazeera.com/news/2022/6/23/algeria-niger-nigeria-resume-talks-on-trans-sahara-gas-pipeline>.

Table 2 Implementation of corporate governance initiatives in oil and pharmaceutical industries

Key elements of corporate governance	How oil/energy industry engaged these elements	How pharmaceutical industry engaged these elements
Stakeholder interests	Transparent sustainability disclosures through publishing comprehensive sustainability reports to reassure investors [95] Localised socio-economic investments through funding community infrastructure projects to secure goodwill [97, 98]	Equitable global access pricing through implementation of tiered pricing schemes to improve drug accessibility [96] Strategic patient advocacy involvements through Funding independent patient organizations to amplify consumer voices [50]
Transparency and disclosure	Standardization of financial and anti-corruption disclosures through publishing regional extraction payments to curb localised corruption [99] Standardized carbon and energy accounting through disclosing carbon emissions metrics under government mandates [101]	Opening public clinical trial registers where all trial data are registered before participant recruitments [100] Health professional financial disclosures which allow for reporting corporate financial transfers to medical practitioners [102]
Legal and ethical compliance	Adoption of integrated project ethical frameworks via embedding integrity parameters into industrial project lifecycles [103] Formal compliance and refinery training via running structured compliance simulations for industrial workforce	Use self-regulatory marketing codes to restrict promotional tactics through universal codes [104] Contract lifecycle management automation allowing for deployment of digital tracing tools that ensure drug chain safety [105]
Accountability and responsibility	Contractual renegotiation of climate contracts by accepting legally binding frameworks for climate damage remediation [106] Localised environmental grievance mechanisms by empowering regional oversight bodies to track drilling pollution [108]	Human rights independent health benchmarking via the adoption of external monitoring panels to evaluate medicine quality [107] Standardized global health benchmarking via participation in competitive corporate indices that prioritize public health [109]
Internal control	Financial and asset fraud prevention via mitigating financial anomalies through integrated oversight tracking [50] Structured risk assessment architecture by deploying framework components to ensure stable efficiency [111]	Inventory integrity and safety management via the optimization of medical inventory control to prevent expiration hazards [110] Good manufacturing practice quality governance via executing independent audit systems validating manufacturing parameters [112]
Management oversight	Dedicated board sustainability committee established to direct emissions strategies [113] Specialised risk management committees to oversee project resource allocation [115]	Setting up separate risk and compliance sub-committees to manage mission-critical vulnerabilities [114] Institutional knowledge management documentation to eliminate executive bias [116][135]

4.3 Cross-case analysis: putting the pieces together

The within-case assessment of the oil and pharmaceutical industries demonstrates that both sectors have integrated corporate governance and social responsibility (CG/SR) practices into their operations. Drawing on key corporate governance principles identified by Lawless [8], this study comparatively examines the components, similarities, and operational mechanisms underlying CG/SR engagement across both industries. These practices represent efforts to strengthen organisational regulation, processes, and decision-making to improve service delivery and societal well-being. The selected elements also enhance the conceptual and practical understanding of AI ethics and governance in the real-world context. Table 2 summarises the CG/SR elements and their application across the two industry groups.

In Table 2, we highlight the key components that facilitate good corporate governance and social responsibility practices across the oil and pharmaceutical industries.

Specifically, it could be seen that Oil/Energy companies engage with and promote stakeholders' interests through transparent sustainability disclosures and localised socio-economic investments while the pharmaceutical industry actively manages and promotes stakeholder interests through targeted access and patient-centric programs. Legal and ethical compliance monitoring helps both sectors mitigate liabilities, protect consumers, and operationalise institutional ethics. Corporate accountability and responsibility mechanisms are heavily scrutinized in both high-impact sectors. Companies actively leverage specific operational and external frameworks to justify their actions to the public, regulators, and legal authorities. Internal control systems are vital for protecting assets, preventing fraud, and ensuring operational reliability in both high-risk sectors. Management oversight mechanisms ensure that boards of directors effectively monitor executive decision-making, manage structural risks, and protect shareholder assets within heavily regulated environments.

In subsequent sections, we discuss the findings from this analysis and their applications and implications to AI governance and responsible technology innovations in the real-world.

5 Reflexive discussions and broader implications for practice

A reflexive discussion is an ongoing process that allows researchers to critically examine their biases, assumptions and positionality—shifting the focus to appreciating how the background of the researchers shape the understanding of the study [117–119].

Authors background This article presents our interdisciplinary perspectives on current corporate governance and social responsibility (CG/SR) practices, challenges, and paradoxes in the oil/energy and pharmaceutical industries. Drawing on our diverse international backgrounds and research experiences, we offer insights shaped by extensive engagement with these sectors.

- The first author is a Senior Lecturer in Computer Science at University of Greenwich, London and co-founder of Responsible AI Across Borders. As an interdisciplinary researcher with experience in Higher Education across the UK and Africa, he explores human-computer interaction, AI ethics and governance, and persuasive computing, developing human-centered and accessible sociotechnical systems. Through academic leadership, he advances SDGs, inspires future researchers, and drives impactful tech solutions.
- The second author is a Research Fellow in Responsible AI for Health at University of Nottingham and co-founder of Responsible AI Across Borders. His research advances the “AI in/from Africa” agenda through responsible research and innovation, health data governance, and responsible digital futures, with strong contributions to African perspectives on human-computer interaction and AI.
- The third author is a Senior Lecturer in Pharmaceutical Sciences at University of Greenwich, London, specialising in mucoadhesive drug delivery and AI-driven pharmaceutical engineering. His research combines rheology, texture analysis, and machine learning for formulation optimisation. He supervises MSc/PhD students, serves as Associate Editor for the the Journal of The Royal Society Interface, and advances pharmaceutical innovation through research and entrepreneurship.
- The fourth author is an AI Researcher, Strategist, and Policy Advisor, and Founder and Scientific Director of Technocultura, focusing on AI governance and safety for the Global Majority. She earned her PhD in Computer Science in 2023, serves as Policy Lead for Black in AI, and has advised the UN, World Bank, and other leading organisations.
- The fifth author is a Canada Research Chair in Persuasive Technology and Professor of Computer Science at Dalhousie University Canada, directing the Persuasive Computing Lab. Appointed by the United Nations General Assembly and Secretary-General to its first global independent scientific body on AI, her research explores human-centred technologies that promote social good, improving health, safety, sustainability, and positive behavioural change through interactive systems.

As interdisciplinary researchers in responsible AI, human-computer interaction, social computing, and pharmaceutical science, we believe that a critical analysis of key components associated with the corporate governance and social responsibility (CG/SR) initiatives in the oil and pharmaceutical industries will provide valuable lessons for the development of more inclusive, context-sensitive, and effective AI ethics and governance strategies. In addition, we hope that this article will inspire a dynamic approach to AI governance policy design and processes which prioritize safety and reliability over quick deployments, offer opportunities for distributed public private AI testing and validation protocols, and emphasize shared cross-border recognition, adoption and dynamic implementation of integrated AI governance frameworks.

This section is organised in two parts. First, it reviews the challenges/pressures (subsection 5.1) and second, the paradoxes (subsection 5.2) that influence how corporate governance and social responsibility processes are understood/applied in the oil and pharmaceutical industries, and their implications to AI ethics and governance works across stakeholders including researchers and designers, practitioners and deployers, government and policymakers, as well as advocacy/right groups.

5.1 Corporate governance challenges in oil & pharmaceutical sectors: AI governance implications

Here, we present the challenges of corporate governance and social responsibility in the oil and pharmaceutical industries, and their implications to AI governance and future technology governance in general.

Strict regulatory and accountability frameworks As identified in Table 2, governments have increasingly shifted from voluntary corporate responsibility models towards legally mandated transparency and accountability frameworks, supported by significant financial penalties. For example [120], demonstrated that integrating “human rights due diligence” into international trade protocols can compel pharmaceutical executives to address public welfare failures while promoting accessibility, availability, acceptability, and fairness in healthcare. These insights have important implications for AI ethics and governance. Similar regulatory approaches can move AI governance from reactive responses to proactive accountability by requiring organisations to embed ethical and legal compliance into AI systems by design. For AI and technology companies, emerging national and international regulations (see Table 1) will require transparent, auditable, and robust documentation of algorithmic safety, security, and risk management practices before deployment. This approach will support responsible innovation and strengthen public trust in AI technologies.

Unpredictable institutional investor and ESG challenges Capital markets increasingly tie investment decisions to organisations’ Environmental, Social, and Governance (ESG) performance, compelling boards to strengthen governance and risk management as identified in Table 2. A global study of 264 oil and gas firms found that severe ESG controversies significantly reduce market valuation, increase corporate risk, and threaten long-term sustainability [121]. These findings are equally relevant to AI ethics and governance. Boards of AI and Big Tech companies must recognise that AI governance is not merely a public relations concern but a fundamental business responsibility. They should treat computational sustainability, including the energy demands and carbon footprint of AI infrastructure, and algorithmic bias as material governance and compliance risks. Failure to address these issues can undermine investor confidence, damage organisational reputation, reduce market value, and limit access to capital, making responsible AI governance essential for long-term competitiveness and resilience.

Heightened civil society awareness and NGO mobilisation Activists, host communities, independent researchers, and non-governmental organisations can directly threaten a firm’s operational sustainability by exposing ethical failures and pursuing legal action over practices that harm people and the environment as identified in Table 2. For example, Alomar et al. (2025) found that NGO-driven public pressure is a key catalyst compelling firms in the oil and energy sector to improve their environmental performance, alongside government influence and other stakeholder forces

[122]. This finding has important implications for AI ethics and governance. The growing influence of digital rights organisations, open-source communities, and whistleblowers has increased scrutiny of data privacy and algorithmic harms. Similarly, national and international AI rights groups can pressure AI and Big Tech firms to address discriminatory algorithms and other adverse impacts on marginalised communities. Such external pressure is likely to encourage organisations to establish transparent public grievance mechanisms and adopt independent, participatory algorithmic auditing practices to strengthen accountability, maintain public trust, and protect their reputations.

5.2 Governance paradoxes in oil and pharmaceutical sectors: implications for AI governance

Here, we present the contradictions in the operations of oil and pharmaceutical industries, and their implications to AI governance in particular and emerging technology governance in general.

Shareholder profit expansion versus broader social responsibility initiatives Generally, corporate law requires firms to maximise shareholder value. However, this obligation can conflict with the substantial investment needed to improve environmental, social, and governance (ESG) performance and support corporate social responsibility (CSR) initiatives as identified in Table 2. A mixed-methods study of CSR in the pharmaceutical sector found that decisions prioritising public benefit often reduce short-term profits [50]. Similar tensions are evident in AI and Tech companies, where commercial incentives to monetise user data frequently conflict with privacy and data protection. We argue that AI governance should adopt more adaptive corporate structures to balance financial and societal objectives. One example is Anthropic’s Public Benefit Corporation (PBC) model, which legally requires companies to pursue both shareholder returns and public benefit. Another is OpenAI’s proposal to grant the United States government a 5% equity stake⁷ to ease political concerns surrounding AI governance. According to CEO Sam Altman, this would help share AI’s benefits and responsibilities with the public [123]. While such models may protect ethical AI practices from purely profit-driven shareholder litigation, they may also weaken accountability, potentially enabling data privacy breaches,

⁷ <https://www.ft.com/content/7c803eab-8e80-4431-9a87-e943bf00e00b?syn-25a6b1a6=1>.

security risks, and human rights violations if governance and regulatory oversight remain inadequate.

Short-term business demands versus long-term sustainable innovation Equity markets demand predictable quarterly earnings, but addressing systemic risks would need high-risk, longer-term capital perspectives as identified in Table 2. For instance, Halttunen et al. [124] examined the pressure on companies to contribute to the social goal of climate change mitigation versus the need to perform financially and meet obligations to shareholders in activities that directly contribute to climate change. The results reveal a structural paradox where corporate managers struggle to reconcile legacy business models with long-term survival under existential constraints [124]. This clearly maps to the current AI race issues, where AI firms take unfinished generative AI models to the market to satisfy quarterly investor publicity, despite massive long-term risks regarding data bias, cybersecurity vulnerabilities, misinformation, and copyright infringement. AI and indeed BigTech governance initiatives must consider this contradiction and explore strategic ways to integrate checkpoints/guardrails that will ensure long-term systemic safety and reliability takes precedence over short-term feature deployments and perceived benefits.

Superficial marketing alignment versus core operational realities While governance structures may establish sustainability committees to drive emissions strategies as identified in Table 2, sometimes, they may sponsor outward-facing public relations campaigns that directly contradict the environmental or systemic harm caused by their primary business model. An existing study highlights how corporate structures fall into “green capitalism” traps, treating sustainability as a promotional gimmick rather than re-engineering destructive practices that will promote peoples’ wellbeing and safe environment [125]. This surfaces moral issues of ethics washing [38], where AI and other Big-Tech firms would publish sophisticated “Responsible AI Principles” while simultaneously downsizing their internal trust, safety, and algorithmic fairness teams to cut costs. This clearly highlights the need to separate ethics teams from marketing departments, giving compliance officers independent veto power over product launches to ensure core operational practices align with public declarations.

In the subsequent section, this paper offers a shift in thinking - proposing an interdisciplinary framework for reframing and enhancing AI governance. While this enhanced framework integrates the findings from our cross case analysis, it also presents an inclusive and sustainable AI governance as an evolving socio-technical development that emphasises

stakeholder participation, interdisciplinary perspectives, safety and trust over feature deployment, responsive adjustment and contextual sensitivity to real-world and cross-border political, cultural, and environmental underlying forces.

6 Interdisciplinary contributions for reframing and enhancing AI governance frameworks

Current AI governance paradigms, including strategies, laws, frameworks, and standards, promote ethical principles, legal compliance, and safety while protecting human rights and discouraging system manipulation through enforceable regulations [126, 127]. However, these mechanisms can create organisational rigidity that delays AI deployment. Although designed to improve governance efficiency and support large-scale data auditing, they often overlook tacit qualitative knowledge and may conflict with privacy requirements because of permanent digital records [128]. While large technology corporations incorporate multi-stakeholder perspectives into AI policy development, governance often remains voluntary, lacking binding commitments and leaving organisations vulnerable to ethics washing. Existing human-centred AI research also suggests that sociotechnical approaches inadequately address contextual social, political, and environmental influences [46, 47]. Consequently, these governance limitations increase the risk of biased algorithmic outcomes, privacy and security concerns, and weak accountability. They also encourage organisations to rely on reactive corporate responses that diminish transparency and political responsibility, particularly for historically marginalised communities disproportionately affected by AI systems.

Our research paper addresses the above weaknesses. We highlight four (4) critical components/insights that could be used to enhance/reframe AI ethics and governance from an interdisciplinary perspective, and which is grounded in the findings from a sector-level critical synthesis of corporate governance and social responsibility practices in the oil and pharmaceutical industries sectors. While they are not AI governance frameworks on their own, these components will inform real-world applications and policy implications on not only AI governance but also emerging technology governance across geographical borders. They are:

1. From siloed to dynamic policy processes: Beyond enacting isolated national, international and corporate AI strategies (distributed governance of AI), proprietors of AI governance frameworks should also consider enabling interoperable regulatory and accountability processes that are grounded in cognate interdisciplinary

perspectives. This can be achieved by facilitating participatory AI governance meetings (among national, international and corporate institutions) that are policy-focused and designed to identify borderlines and streamline implementation protocols across boards. The resulting AI policies and processes would be inclusive, advancing stakeholder's objectives.

2. Safety and reliability over deployments: AI governance should emphasise an ethics-by-design paradigm where ethics principles such as safety, trust and reliability of algorithmic systems are prioritized over feature deployments and product benefits. This can be achieved by decoupling trust and safety verification layers from product-marketing motivations and allowing independent and objective verification and validations of algorithmic systems across the various phases of the AI system lifecycle.
3. Agile testing environments: Increasing awareness and rapid emergence of national and international AI rights groups demands the urgent deployment of public-private testing environments to validate AI system stability, as well as ethical, legal and safety compliance. In practice, an agile testing environment (consisting of dedicated hardware, software, and network setup used to evaluate software under controlled conditions) would allow multidisciplinary teams to run AI test cases, identify ethical issues, and co-creatively develop and validate fixes without impacting end-users of AI systems.
4. Shared cross-border recognition: This involves a shared responsibility amongst stakeholders including national, regional and organisational policymakers in designing strategies to support the adoption and dynamic implementation of integrated AI governance frameworks. This will allow for standard accountability and safety audits to be recognised and supported across geographic boundaries and jurisdictions.

Essentially, our research contribution enhances existing frameworks by acknowledging that AI governance is an evolving socio-technical process that should incorporate dynamic and interdisciplinary perspectives, enabling agile adaptation to emerging ethical and governance problems across diverse contexts [129]. It emphasises safety, trust and reliability of algorithmic systems over feature deployments and product marketing benefits [130, 131]. While our enhanced AI governance framing facilitates participatory governance of AI systems, it also supports responsive adjustment and contextual sensitivity to real-world and cross-border political, cultural, and environmental underlying forces [132], preventing AI and emerging Big-Tech companies from exploiting cross border gaps in regulatory policies and governance.

7 The way forward: operationalising interdisciplinary framework for AI governance

In this section, we present some ideas to operationalise our AI governance framing. Essentially, everyone stakeholder, including researchers and designers, practitioners and deployers, government and policymakers, as well as advocacy/right groups would play important roles in realising the objectives of this innovative framing for AI ethics and governance in the real-world.

1. Researchers and designers: Research and design teams can establish theoretical and conceptual foundations for dynamic regulatory and accountability processes informed by interdisciplinary perspectives. They can advance the development of real-time testing environments and open-source systems that enable public-private partnerships to analyse and validate algorithmic bias, data safety, human rights protection, regulatory compliance, and computational energy use without relying solely on corporate inputs. Similar to corporate governance and social responsibility initiatives in the oil and pharmaceutical sectors, these collaborative approaches encourage shared responsibility among stakeholders. Furthermore, shifting from product-focused development toward ethics-by-design principles, including safety-by-design and trust-by-design, allows AI product developers to embed human-centred safeguards and accountability mechanisms within software architectures. This approach generates empirical evidence needed to assess system efficiency, effectiveness, and ethical performance throughout the AI lifecycle, from development to post-deployment.
2. Practitioners and deployers: Practitioners and deployers will convert the theoretical and conceptual foundation (created by researchers and designers) into technical workflows, leading to commercialisation and deployment. The revamping of existing internal guardrails within the workflow of these AI/BigTech companies using relevant components from our new AI governance framing, could ensure that ethical concerns; ethics-by-design (safety-by-design, trust-by-design, etc.) are prioritised over rapid feature deployments and product benefits. In line with guidelines for participatory (co-design) methodologies [133, 134], practitioners and deployers alongside regulatory agencies and community groups would be actively involved in the agile testing environments, where they will support with sharing safe data handling practices and help with sorting technical issues as they occur during the public-private feature testing exercises. This partnership bridges the

gap between high-level policy and real-world system capabilities, ensuring that compliance practices remain dynamic, reliable, and effective under market pressures.

3. Government and policymakers: The government and policymakers are saddled with the responsibility of creating the legal framework (including regulatory bodies) to manage and scale this inclusive AI governance strategies and penalise non-compliance. Beyond the current fragmented national, international and organisational AI strategies, policymakers could drive dynamic and integrated global agreements that align AI policies and regulations with shared priorities. This kind of integrated structuring of AI governance is timely and strategic and has the potential to prevent AI and emerging tech companies from exploiting regulatory gaps across borders, especially across marginalised communities in the global south. Additionally, they could fund the establishment of agile testing labs across their respective jurisdictions to ensure that AI systems deployed in their communities undergo social and safety examinations.
4. AI right groups and non-governmental organisations: As external ethical actors, emerging national and international AI rights groups and civil society organisations can ensure that public welfare and citizen participation remain central to co-creating inclusive and sustainable AI futures. Through their mobilisation networks and interdisciplinary expertise, they can advocate for corporate governance and social responsibility practices, similar to those developed in the oil and pharmaceutical sectors, to be adapted for AI. These groups can strengthen corporate accountability, challenge algorithmic discrimination, and expose ethics-washing practices. They can also contribute to public-private partnerships by participating in agile AI system testing to assess ethical, legal, and safety compliance. By capturing and analysing user experiences, they can help translate social concerns into evolving governance requirements, ensuring transparent communication between AI developers, governments, and marginalised communities while promoting fair and responsible AI innovation.

8 Conclusion, limitation and future work

This study employed a multi-case analysis to examine corporate governance and social responsibility practices in the oil and pharmaceutical industries, identifying six key components that support effective governance across both sectors. First, the oil companies engage with and promote stakeholders' interests through transparent sustainability disclosures and localised socioeconomic investments, while the pharmaceutical industry actively manages and promotes

stakeholder interests through targeted access and patient-centric programs. Second, legal and ethical compliance monitoring helps both sectors mitigate liabilities, protect consumers, and operationalise institutional ethics. Third, corporate accountability and responsibility mechanisms are heavily scrutinised in both high-impact sectors. Fourth, companies actively leverage specific operational and external frameworks to justify their actions to the public, regulators, and legal authorities. Fifth, internal control systems are vital for protecting assets, preventing fraud, and ensuring operational reliability in both high-risk sectors. Finally, management oversight mechanisms ensure that boards of directors effectively monitor executive decision-making, manage structural risks, and protect shareholder assets within heavily regulated environments. The study also identified challenges and paradoxes associated with corporate governance and social responsibility practices, with important implications for AI governance. Lessons from these sectors contribute to global discussions on responsible AI research, ethics, and governance by demonstrating how accountability, transparency, and stakeholder-centred approaches can inform emerging AI frameworks.

Based on the findings, this article produced four (4) conceptual research insights that could inform real-world applications and policy implications on not only AI governance but also emerging technology governance across geographical borders. The key components of the research insights are: (1) Enablement of dynamic regulatory and accountability processes that are grounded in cognate interdisciplinary perspectives. (2) Emphasize ethics by-design paradigm where ethics principles such as safety, trust and reliability of algorithmic systems are prioritised over feature deployments and product benefits. (3) Demands for urgent deployment of public-private testing environments to validate AI system stability, as well as ethical, legal and safety compliance. (4) Implementation of integrated AI governance frameworks that allow for standard accountability and safety audits to be recognised across national and international borders. The study also recommends bespoke ideas for putting the AI governance framework to work in the real-world.

8.1 Limitation

The multi-case analysis effectively captures context-specific insights; however, this study relied on critical analysis of literature and reports from diverse sources rather than direct stakeholder engagement across company locations, which would require substantial resources for travel, interviews, and data management. This study conducted sector-level critical synthesis of corporate governance initiatives, with the oil and pharmaceutical industries serving as source domains. However, an abridged discussion was presented

instead of extensive within case analysis. This is because of article page limitations. The authors adopted a purposive case selection of two industries based on our conceptual knowledge and research experience. Both industries were selected because they are deeply related through shared high-stakes risks controls and long history of established corporate governance practices that has witnessed intense public scrutiny over the years and across geographical contexts. However, because they were randomly selected, the findings may not be statistically generalisable. Additionally, cross-case comparisons may involve researcher bias. However, the interdisciplinary expertise of the research team in responsible AI, human-computer interaction, pharmaceutical sciences, responsible innovation, and behavioural modelling supported rigorous and balanced analysis.

8.2 Future work

In the future, we plan to collaborate with interested stakeholders across the national, international and organisational AI policy makers to integrate our suggestions to their AI strategies. In addition, our research will involve fieldwork across oil and pharmaceutical operations in selected low-, middle-, and high-income countries, engaging stakeholders, including NGOs, civil society, and host communities. By capturing lived experiences, we will evaluate the real-world impact of corporate governance and social responsibility initiatives, generating evidence to inform more effective, and sustainable AI governance strategies that are inclusive by design.

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Data availability No datasets were generated or analysed during the current study.

Declarations

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References

1. United Nations: Global Dialogue on AI Governance. United Nations (2026). <https://www.un.org/global-dialogue-ai-governance/en>
2. Gwagwa, A., Kazim, E., Kachidza, P., Hilliard, A., Siminyu, K., Smith, M., Shawe-Taylor, J.: Road map for research on responsible artificial intelligence for development (AI4D) in African countries: the case study of agriculture. *Patterns*. **2**(12), 100381 (2021)
3. Eitel-Porter, R.: Beyond the promise: implementing ethical AI. *AI Ethics*. **1**(1), 73–80 (2021)
4. Hagendorff, T.: The ethics of AI ethics: an evaluation of guidelines. *Mind. Mach.* **30**(1), 99–120 (2020)
5. Van Wynsberghe, A.: Sustainable AI: AI for sustainability and the sustainability of AI. *AI Ethics*. **1**(3), 213–218 (2021)
6. Li, B., Qi, P., Liu, B., Di, S., Liu, J., Pei, J., Zhou, B.: Trustworthy AI: from principles to practices. *ACM Comput. Surveys*. **55**(9), 1–46 (2023)
7. Floridi, L.: Translating principles into practices of digital ethics: five risks of being unethical. *Philos. Technol.* **32**(2), 185–193 (2019)
8. Lawless, J.: What is the relationship between corporate governance and social responsibility? Purple Griffon. Retrieved June 11, 2026, from (2024). <https://purplegriffon.com/blog/relationship-between-corporate-governance-and-social-responsibility>
9. Nwankwo, E., Sonna, B.: Africa's social contract with AI. *XRDS: Crossroads ACM Magazine Students*. **26**(2), 44–48 (2019). <https://doi.org/10.1145/3368073>
10. Reyes-Cruz, G., Vallejos, E. P., Barnard, P., Schneiders, E., Tachiquin, M., Price, D., Fischer, J. Unravelling responsible AI: an umbrella review. In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* (Vol. 8, No. 3, pp. 2177–2188)
11. Onatu, G.: The Use Of Case Study Research Methodology and Design as Research Tool, *INTED2012 Proceedings*, pp. 3946–3953. (2012)
12. Greenhalgh, T.: Case studies: a guide for researchers, educators, and implementers. *BMJ Med.* **4**(1), e001623 (2025). <https://doi.org/10.1136/bmjmed-2025-001623>
13. National University Library. (n.d.). Case study. National University (2026). Retrieved June 11: from <https://resources.nu.edu/researchtools/casestudy>
14. Cochrane Information Specialists' Handbook Section: 5. <https://training.cochrane.org/resource/tscinduction-mentoring-training-guide/5-handsearching>. Retrieved 09/04/2025
15. Dickersin, K., Scherer, R., Lefebvre, C.: Identifying relevant studies for systematic reviews. *BMJ*. **309**(6964), 1286–1291 (1994). <https://doi.org/10.1136/bmj.309.6964.1286>
16. Ibitoye, A.O., Nkwo, M.S., Orji, R.: Rethinking responsible AI from ethical pillars to sociotechnical practice. *AI Ethics*. **5**, 6207–6223 (2025). <https://doi.org/10.1007/s43681-025-00809-2>
17. Cao, L.: AI in finance: challenges, techniques, and opportunities. *ACM Comput. Surv. (CSUR)*. **55**(3), 1–38 (2022). <https://doi.org/10.1145/3502289>

18. Mohammed, Y.S.: Applications of artificial intelligence (AI) in healthcare: a review. *ScienceOpen Preprints*. (2021). <https://doi.org/10.14293/S2199-1006.1.SOR-PPVRY8K.v1>
19. Deshpande, A., Sharp, H.: Responsible AI systems: who are the stakeholders? In: *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 227–236). (2022)
20. European Commission: AI Act. (2025). <https://digital-strategy.ec.europa.eu/en/policies/regulatoryframework-ai>
21. OECD.AI. OECD.AI Policy Navigator. Organisation for economic co-operation and development. Retrieved June 9, 2026, from (2025). <https://oecd.ai/en/dashboards/overview>
22. Tjondronegoro, D., Yuwono, E., Richards, B., Green, D.: Responsible AI implementation. a human-centered framework for accelerating the innovation process (2022)
23. Schiff, D., Rakova, B., Ayes, A., Fanti, A., Lennon, M.: Principles to practices for responsible AI: Closing the Gap (2020). *ArXiv*. <https://arxiv.org/abs/2006.04707>
24. Dignum, V.: Responsibility and artificial intelligence. *Oxford Handbook Ethics AI* **4698**, 215 (2020)
25. Shneiderman, B.: Responsible AI: Bridging from ethics to practice. *Commun. ACM*. **64**(8), 32–35 (2021)
26. Ribeiro, D., Rocha, T., Pinto, G., Cartaxo, B., Amaral, M., Davila, N., Camargo, A.: Toward effective AI governance: a review of principles (2025). (arXiv:2505.23417). *arXiv*. <https://doi.org/10.48550/arXiv.2505.23417>
27. Butcher, J., Beridze, I.: What is the state of artificial intelligence governance globally? *RUSI J.* **164**(5–6), 88–96 (2019)
28. Schmitt, L.: Mapping global AI governance: a nascent regime in a fragmented landscape. *AI Ethics*. **2**(2), 303–314 (2022)
29. Bullock, J.B., Chen, Y.C., Himmelreich, J., Hudson, V.M., Korinek, A., Young, M.M., Zhang, B. (eds.): *The Oxford handbook of AI governance*. Oxford University Press (2024)
30. Diallo, K., Smith, J., Okolo, C.T., Nyamwaya, D., Kgomo, J., Ngamita, R.: Case studies of AI policy development in Africa. (2024). *arXiv preprint arXiv:2403.14662*.
31. International Organization for Standardization ISO: ISO/IEC 42001:2023 information technology— artificial intelligence—management system. ISO. (2023). <https://www.iso.org/standard/42001>
32. National Institute of Standards and Technology (NIST): AI risk management framework (AI RMF) (2024). <https://www.nist.gov/itl/ai-risk-management-framework>
33. Department for Digital, Culture, Media and Sport: National AI strategy. UK Government (2021). https://assets.publishing.service.gov.uk/media/614db4d1e90e077a2cbdf3c4/National_AI_Strategy_PDF_version.pdf
34. Okolo, C.T.: AI in the Global South: Opportunities and challenges towards more inclusive governance, vol. 2. The Brookings Institution (2023)
35. Cihon, P., Maas, M.M., Kemp, L.: Fragmentation and the future: Investigating architectures for international AI governance. *Global Policy*. **11**(5), 545–556 (2020). <https://doi.org/10.1111/1758-5899.1289>
36. Hassan, Y.: Governing algorithms from the South: a case study of AI development in Africa. *AI Soc.* **38**, 1429–1442 (2023). <https://doi.org/10.1007/s00146-022-01527-7>
37. Ogoh, G., Eke, D., Stahl, B.C.: AI governance in North Africa: an ecosystems approach. *Sci. Public. Policy*. **53**(2), 258–276 (2026). <https://doi.org/10.1093/scipol/scag007>
38. Nkwo, M., Adamu, M.: AI Ethics shopping and governance shrinking in Africa: a critical opinion. *Greenwich Academic Literature Archive* (2024). https://gala.gre.ac.uk/id/eprint/47630/7/47630%20NKWO_AI_Ethics_Shopping_And_Governance_Shri%20ng_In_Africa_%28Draft%29_2024.pdf
39. Effoduh, J.O.: Decolonizing the governance of artificial intelligence in Africa: from normative mimicry to epistemic sovereignty. *Sci. Public. Policy*. **53**(2), 245–257 (2026). <https://doi.org/10.1093/scipol/scag005>
40. Smuha, N.A.: From a ‘race to AI’ to a ‘race to AI regulation’: regulatory competition for artificial intelligence. *Law Innov. Technol.* **13**(1), 57–84 (2021). <https://doi.org/10.1080/17579961.2021.1898300>
41. Salaam, C., Rawat, D., Monroe-White, T.: Towards Holistic AI Governance in Africa. In: Onyango, G. (ed.) *Designing artificial intelligence for public policy and governance in Africa. Studies in national governance and emerging technologies*. Palgrave Macmillan, Cham (2025). https://doi.org/10.1007/978-3-031-83445-5_2
42. Okolo, C.T., Raji, M.: The global majority in international AI governance. *arXiv preprint arXiv:2601.17191*. (2026)
43. Birhane, A., Angius, R., Agnew, W., Pandit, H.J., Mitra, B., Dobbe, R., Talat, Z.: Big AI’s Regulatory Capture: Mapping Industry Interference and Government Complicity. In: *9th ACM Conference on Fairness, Accountability, and Transparency* (pp. 2586–2607). (2026). <https://doi.org/10.1145/3805689.3806740>
44. Ishkhanyan, A.: Governing AI across borders: corporate power, state sovereignty and global regulation. *Digit. Policy Regul. Gov.* **28**(3), 280–295 (2026). <https://doi.org/10.1108/DPRG-032025-0057>
45. Dafoe, A.: AI governance: a research agenda. *Governance of AI Program, Future of Humanity Institute*. Oxford, UK, University of Oxford (2018)
46. Adamu, M., Nkwo, M.: AI in Africa: Preliminary notes on design and adoption. *diid disegno industriale industrial Des.* **80**, 44–57 (2023).
47. Eke, D.O., Wakunuma, K., Akintoye, S.: Introducing responsible AI in Africa. In: Eke, D.O., Wakunuma, K., Akintoye, S. (eds.) *Responsible AI in Africa. Social and Cultural Studies of Robots and AI*. Palgrave Macmillan, Cham (2023). https://doi.org/10.1007/978-3-031-08215-3_1
48. Robinson, M.: OGP and EITI: a shared agenda on anti-corruption and the energy transition. *Extractive Industries Transparency Initiative* (2023). <https://eiti.org/blog-post/ogp-and-eiti-sharedagenda-anti-corruption-and-energy-transition>
49. Nwankwo, S.E.-O.T.C.: Christiana Adanma Otonnah and Ekene Ezinwa Nwankwo. Corporate governance and CSR for sustainability in oil and gas: trends, challenges, and best practices: a review. *World J. Adv. Res. Reviews*. **21**(3), 078–090 (2024). Article DOI <https://doi.org/10.30574/wjarr.2024.21.3.0662>
50. Wong, A., Saeed, G., Garner, S., Kohler, J.: Evaluating the corporate social responsibility agenda for high-cost novel therapies: roles for government and civil society. *Health Res. Policy Syst.* **23**(1), 157 (2025). <https://doi.org/10.1186/s12961-025-01421-w>
51. Adriana Silva, C.A.N., Proença, D., Maria Elisabete Neves: Governance and management drivers of ESG-washing in the pharmaceutical industry: an empirical analysis. *Social Responsib. J.* (2026). Advance online publication <https://doi.org/10.1108/SR-J-05-20250499>
52. Hopewell, S., Clarke, M., Lefebvre, C., Scherer, R.: Handsearching versus electronic searching to identify reports of randomized trials. *Cochrane Database Syst. Rev.* **2007**(2), MR000001 (2007)
53. Vassar, M., Atakpo, P., Kash, M.J.: Manual search approaches used by systematic reviewers in dermatology. *J. Med. Libr. Assoc.* **104**(4), 302–304 (2016). <https://doi.org/10.3163/1536-5050.104.4.009>
54. American Chemical Society: Development of the Pennsylvania Oil Industry - National Historic Chemical Landmark - American Chemical Society. (2009). <http://www.acs.org/content/acs/en/education/whatischemistry/landmarks/pennsylvaniaoilindustry>
55. Giddens, P.H.: One hundred years of petroleum history. *Arizona West*. **4**(2), 127–144 (1962). <http://www.jstor.org/Stable/40167748>

56. Koehn, N.F., Miller, K.: John D. Rockefeller and the creation of standard oil—compilation. Harvard Business School Compilation 807 – 110, February 2007. (Revised March 2012.). (2012). <https://www.hbs.edu/faculty/Pages/item.aspx?num=34079>
57. Fenske, J., Zurimendi, I.: Oil and ethnic inequality in Nigeria. *J. Econ. Growth*. **22**(4), 397–420 (2017). <https://doi.org/10.1007/S10887-017-9149-8/TABLES/5>
58. Ogbuigwe, A.: Refining in Nigeria: history, challenges and prospects. *Appl. Petrochemical Res.* 2018. **8**(4), 181–192 (2018). <https://doi.org/10.1007/S13203-018-0211-Z>
59. OPEC. (n.d.). OPEC: Member countries. Retrieved June 28: from (2023). https://www.opec.org/opec_web/en/about_us/25.htm
60. Akpanika, O.: Technology transfer and the challenges of local content development in the Nigerian oil industry. *Global J. Eng. Res.* **11**(2). (2013)
61. Amadi, A.H., Victor, O., Ayoola, J.O.: Review of Nigeria’s petroleum industry bill (PIB). *Eur. J. Eng. Res. Sci.* **5**(9), 1081 (2021)
62. Cordes, E.E., Jones, D.O.B., Schlacher, T.A., Amon, D.J., Bernardino, A.F., Brooke, S., Carney, R., DeLeo, D.M., Dunlop, K.M., Escobar-Briones, E.G., Gates, A.R., Génio, L., Gobin, J., Henry, L.A., Herrera, S., Hoyt, S., Joye, M., Kark, S., Mestre, N.C., Witte, U.: Environmental impacts of the deep-water oil and gas industry: a review to guide management strategies. *Front. Environ. Sci.* **4** (2016). <https://doi.org/10.3389/fenvs.2016.00058>
63. Ogolo, N., Anih, O.C., Onyekonwu, M.O.: Sources and effects of environmental pollution from oil and gas industrial operations. *Arab. J. Chem. Environ. Res.* **90**, 98–121. (2022)
64. Saadoun, I.M.K.: Impact of oil spills on marine life. emerging pollutants in the environment—current and further implications (2015). <https://doi.org/10.5772/60455>
65. Akpan, E.E.: Environmental consequences of oil spills on marine habitats and the mitigating measures—the niger delta perspective. *J. Geoscience Environ. Prot.* **10**(6), 191–203 (2022). <https://doi.org/10.4236/GEP.2022.106012>
66. Heim, I., Vigneau, A.C., Kalyuzhnova, Y.: Environmental and socio-economic policies in oil and gas regions: triple bottom line approach. *Reg. Stud.* **57**(1), 181–195 (2023)
67. Shvarts, E.A., Pakhalov, A., Knizhnikov, A.Y.: Assessment of environmental responsibility of oil and gas companies in Russia: the rating method. *J. Clean. Prod.* **127** (2016). <https://doi.org/10.1016/j.jclepro.2016.04.021>
68. Setzer, J., Higham, C.: Global trends in climate change litigation: 2022 snapshot. (2022). www.ccecep.ac.uk
69. Frynas, J.G.: Corporate social responsibility in the oil and gas sector. *J. World Energy Law Bus.* **2**(3), 178–195 (2009)
70. Spence, D.B.: Corporate social responsibility in the oil and gas industry: the importance of reputational risk (2011). <https://scholarship.kentlaw.iit.edu/cklawreview/vol86/iss1/4>
71. Gielen, D., Boshell, F., Saygin, D., Bazilian, M.D., Wagner, N., Gorini, R.: The role of renewable energy in the global energy transformation. *Energy Strategy Rev.* **24**, 38–50 (2019)
72. Tobin, J.J., Walsh, G.: Medical product regulatory affairs: pharmaceuticals, diagnostics, medical devices. Wiley (2023)
73. Marson, B.M., Concentino, V., Junkert, A.M., Fachi, M.M., Vilhena, R.O., Pontarolo, R.: Validation of analytical methods in a pharmaceutical quality system: an overview focused on HPLC methods, vol. 43, pp. 1190–1203. *Química Nova* (2020)
74. Lucas, S., Ailani, J., Smith, T.R., Abdrabboh, A., Xue, F., Navetta, M.S.: Pharmacovigilance: reporting requirements throughout a product’s lifecycle. *Therapeutic Adv. drug Saf.* **13**, 20420986221125006 (2022)
75. Al-Worafi, Y.M.: Technology for Drug Safety: Current Status and Future Developments. Springer Nature (2023)
76. Davis, A.M., Engkvist, O., Fairclough, R.J., Feierberg, I., Freeman, A., Iyer, P.: Public-private partnerships: compound and data sharing in drug discovery and development. *Slas Discov. Adv. Sci. Drug Discovery*. **26**(5), 604–619 (2021)
77. World Health Organization: Ending the neglect to attain the sustainable development goals: a rationale for continued investment in tackling neglected tropical diseases 2021–2030. World Health Organization (2022)
78. Esfandiari, A., Yazdi-Feyzabadi, V., Zarei, L., Rashidian, A., Salari, H.: Transparency in public pharmaceutical sector: the key informants’ perceptions from a developing country. *BMC Health Serv. Res.* **21**, 1–18 (2021)
79. Garuba, H.A., Kohler, J.C., Huisman, A.M.: Transparency in Nigeria’s public pharmaceutical sector: perceptions from policy makers. *Globalization health.* **5**, 1–13 (2009)
80. Onwujekwe, O., Agwu, P., Odii, A., Orjiakor, C., Obodoechi, D., Nwokolo, C., Roy, P., Hutchinson, E., Mckee, M.: Corruption in the procurement of pharmaceuticals in Anglophone sub-Saharan Africa: a scoping literature review. (2020)
81. Ngozwana, S.: Policies to control prices of medicines: does the South African experience have lessons for other African Countries? Making Medicines in Africa: The Political Economy of Industrializing for Local Health, pp.203–223. (2016)
82. Occa, A., Morgan, S.E., Potter, J.E.: Underrepresentation of Hispanics and other minorities in clinical trials: recruiters’ perspectives. *J. Racial Ethnic Health Disparities.* **5**, 322–332 (2018)
83. Kelsey, M.D., Patrick-Lake, B., Abdulai, R., Broedl, U.C., Brown, A., Cohn, E., Curtis, L.H., Komelasky, C., Mbagwu, M., Mensah, G.A., Mentz, R.J.: Inclusion and diversity in clinical trials: actionable steps to drive lasting change. *Contemp. Clin. Trials.* **116**, 106740 (2022)
84. Lyons, T.: Pharmaceutical Colonialism—Ethical Issues for Clinical Research in Africa. In: Africa in a Restructuring World’International AFSAAP Annual Conference (2009)
85. Okonta, P.I.: Ethics of clinical trials in Nigeria. *Nigerian Med. J. J. Nigeria Med. Assoc.* **55**(3), 188 (2014)
86. Ahmad, K.: Drug company sued over research trial in Nigeria. *Lancet.* **358**(9284), 815 (2001)
87. Lurie, P., Wolfe, S.M.: Unethical trials of interventions to reduce peri-natal transmission of the human immunodeficiency virus in developing countries. *Arguing About Bioeth.* p. 479. (2012)
88. Shah, S.: The Body Hunters: Testing New Drugs on the World’s Poorest Patients. The New Press, New York (2012)
89. Kar, S., Sanderson, H., Roy, K., Benfenati, E., Leszczynski, J.: Green chemistry in the synthesis of pharmaceuticals. *Chem. Rev.* **122**(3), 3637–3710 (2021)
90. Audrine, K., Alfred, V.B.P., Muda, I.: Environmental Stewardship. In: Merck Pharmaceutical Industry. *J. Pharm. Negative Results.* 4114–4120 (2022)
91. Booth, A., Jager, A., Faulkner, S.D., Winchester, C.C., Shaw, S.E.: Pharmaceutical company targets and strategies to address climate change: content analysis of public reports from 20 pharmaceutical companies. *Int. J. Environ. Res. Public Health.* **20**(4), 3206 (2023)
92. Wakui, M.: Corruption and conflicts of interest in the pharmaceutical market: regulation of pharmaceutical companies’ gift-giving practices. In: Competition Law and Policy in the Japanese Pharmaceutical Sector, pp. 69–87. Springer Nature Singapore, Singapore (2022)
93. Pathak, R., Gaur, V., Sankrityayan, H., Gogtay, J.: Tackling counterfeit drugs: the challenges and possibilities. *Pharm. Med.* **37**(4), 281–290 (2023)
94. Al-Hindawi, M.: Marketing authorization and licensing of medicinal products in EU: regulatory aspects. In: *Drug Delivery Trends*, pp. 45–75. Elsevier (2020)
95. Stevens, H., Huys, I.: Innovative approaches to increase access to medicines in developing countries. *Front. Med.* **4**, 218 (2017)

96. Nnaemeka Madumere: The impact of stakeholder management on the oil and gas industry in Africa: a case study of oil companies and African host communities. *African J. Conflict Resol.* **21**, 2 (2022). https://hdl.handle.net/10520/ejc-accordr_v21_n2_a2
97. Doni F., Corvino A., Bianchi Martini S.: Corporate governance model, stakeholder engagementsocial issues evidence from European oilgas industry. *Soc. Responsibility J.* **18**(3), 636–662 (2021)
98. Ajibua, G., Aiyedogbon, J.O.-C.: Impact of corporate social responsibility on the sustainability of the oil and gas industry in Nigeria. *GAS J. Econ. Bus. Manag.* **2**(7), 1–10 (2025). <https://doi.org/10.5281/zenodo.17147948>
99. Parker, L., Mintzes, B.: Patient advocacy organizations and conflicts of interest in research. In: Anderson, E.E. (ed.) *Ethical Issues in Community and Patient Stakeholder–Engaged Health Research. Philosophy and Medicine*, vol. 146. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-40379-8_12
100. Du, J., Zhen, Z., Wei, X., Li, R.: Achieving energy justice through corporate carbon disclosure: the effects of voluntary and third-party verification mechanisms. *J. Bus. Res.* **200**, 115677 (2025)
101. UK Extractive Industries Transparency Initiative: UK EITI guidance for oil and gas companies—2024 process. (2025). <https://www.ukeiti.org/sites/default/files/reports/uk%20eiti%20guidance%20for%20oil%20and%20gas%20%20companies%20-%202024%20process.pdf>
102. Health Research Authority: New research transparency data on UK clinical trials published (2026). <https://www.hra.nhs.uk/about-us/news-updates/new-research-transparency-data-on-uk-clinical-trials-published/>
103. Association of the British Pharmaceutical Industry: Disclosure UK 2024: new record for transparency, with nine in ten healthcare professionals named in database. ABPI Blog (2025)
104. Bakare, O.A., Aziza, O.R., Uzougbo, N.S., Oduro, P.: Ethical and legal project management framework for the oil and gas industry. *Int. J. Appl. Res. Social Sci.* **6**(10), 2363–2388 (2024). <https://doi.org/10.51594/ijarss.v6i10.1618>
105. Francer, J., Izquierdo, J.Z., Music, T., Narsai, K., Nikidis, C., Simmonds, H., Woods, P.: Ethical pharmaceutical promotion and communications worldwide: codes and regulations. *Philos. Ethics Humanit. Med.* **9**, 7 (2014). <https://doi.org/10.1186/1747-5341-9-7>
106. Joseph, I.: Importance of compliance with regulations in the pharmaceutical industry November 30, Available at SSRN: <https://ssrn.com/abstract=4679812> or (2023). <https://doi.org/10.2139/ssrn.4679812>
107. Sholikin, A., Sena, K.M.N.: Promoting transparency and accountability in the oil and gas extractive industry sector for sustainable development in bojonegoro regency. *Jurnal Ilmu Administrasi: Media Pengembangan Ilmu Dan. Praktek Administrasi.* **21**(2), 197–208 (2024). <https://doi.org/10.31113/jia.v21i2.1187>
108. Prasad, M.D., Ansari, S.: Legal and ethical accountability of fossil fuel corporations: need for a new social contract. *J. Clean. Prod.* **423**, 138830 (2023). <https://doi.org/10.1016/j.jclepro.2023.138830>
109. PLoS Medicine Editors: Drug companies should be held more accountable for their human rights responsibilities. *PLoS Med.* **7**(9), e1000344 (2010). <https://doi.org/10.1371/journal.pmed.1000344>
110. Berbash, W., Ektik, M.: Examining the effect of internal control systems on organizational effectiveness: libya oil and gassector study. *Amasya Üniversitesi Ekonomi Ticaret ve Pazarlama Dergisi.* **2**(2), 71–82 (2025)
111. Zakaria, K.M., Nawawi, A., Salin, A.S.A.P.: Internal controls and fraud—empirical evidence from oil and gas company. *J. Financial Crime.* **23**(4), 1154–1168 (2016). <https://doi.org/10.1108/JFC-042016-0021>
112. Sormin, R.N., Manurung, A., Siringoringo, M.J.B.: Analyzing the effectiveness of internal control systems in managing pharmaceutical inventories at Advent Hospital Medan, Indonesia. *Gold. Ratio Data Summary.* **5**(3), 660–667 (2025)
113. Tawade, M., Pandya, S.: Optimizing pharmaceutical quality audits: enhancing compliance and performance through best practices. *J. New. Z. Herpetology* **12**, 6514–6522. (2023)
114. Yahya Jafeel, A., Abdelbagi Abdalla, Y., Amin Abdalla, A., Hersi Warsame, M.: How corporate governance quality affects investment efficiency? An empirical analysis of nonfinancial companies in the Gulf Cooperation Council 2015–2020. *Cogent Bus. Manag.* (2023). <https://doi.org/10.1080/23311975.2023.2198061>
115. Arhinful, R., Mensah, L., Amin, H.I.M., Obeng, H.A., Gyamfi, B.A.: The strategic role of sustainable finance in corporate reputation: a signaling theory perspective. *Sustainability.* **17**(11), 5002 (2025). <https://doi.org/10.3390/su17115002>
116. Katz, D.A., McIntosh, L.A.: Board structure is key to oversight. Harvard Law School Forum on Corporate Governance. (2021). <https://corpgov.law.harvard.edu/2021/09/27/board-structure-is-key-to-oversight/>
117. Olmos-Vega, F.M., Stalmeijer, R.E., Varpio, L., Kahlke, R.: A practical guide to reflexivity in qualitative research: AMEE Guide 149. *Med. Teach.* **45**(3), 241–251 (2023). <https://doi.org/10.1080/0142159X.2022.2057287>
118. Watt, D.: On becoming a qualitative researcher: the value of reflexivity. *Qualitative Rep.* **12**(1), 82–101 (2007). Retrieved from <http://go.unimelb.edu.au/dm56>
119. Jamieson, M.K., Govaart, G.H., Pownall, M.: Reflexivity in quantitative research: a rationale and beginner's guide. *Soc. Pers. Psychol. Compass.* **17**(4), e12735 (2023). <https://doi.org/10.1111/spc3.12735>
120. Lemmens, T., Ghimire, K.M., Perehudoff, K., Kohler, J.C., Low, E., Persaud, N., et al.: Pharmaceutical governance system for costly drugs through human rights due diligence. *BMJ.* **391**, e074982 (2025). <https://doi.org/10.1136/bmj2023-074982>
121. García-Amate, A., Ramírez-Orellana, A., Rojo-Ramírez, A.A., et al.: Do ESG controversies moderate the relationship between CSR and corporate financial performance in oil and gas firms? *Humanit. Soc. Sci. Commun.* **10**, 749 (2023). <https://doi.org/10.1057/s41599-023-02256-y>
122. Alomar, T., Siriwardhane, P., Khan, T.: Contradictory Stakeholder pressures/influences for the oil and gas industry's environmental performance-implications for the current era. *Sustainability.* **17**(24), 11146 (2025). <https://doi.org/10.3390/su17241146>
123. Annie Bao OpenAI proposes US government own 5% stake to address political blowback. CNBC (2026). <https://www.cnbc.com/2026/07/02/openai-proposes-us-government-own-5percent-stake-to-addresspolitical-blowback.html>
124. Halttunen, K., Slade, R., Staffell, I.: We don't want to be the bad guys: oil industry's sensemaking of the sustainability transition paradox. *Energy Res. Social Sci.* **92**, 102800 (2022). <https://doi.org/10.1016/j.erss.2022.102800>
125. GSC Blog Administrator: Green capitalism versus greenwashing: saving the Earth one marketing ploy at a time. *Global Social Challenges Blog* (2025). <https://sites.manchester.ac.uk/global-socialchallenges/2025/01/15/green-capitalism-versus-greenwashing-saving-the-earth-one-marketing-ploy-at-a-time/>
126. Hanisch, M., Goldsby, C.M., Fabian, N.E., Oehmichen, J.: Digital governance: a conceptual framework and research agenda. *J. Bus. Res.* **162**, 113777 (2023). <https://doi.org/10.1016/j.jbusres.2023.113777>
127. Bazoobandi, S., Mahapatra, S., Schipper, T., Wiczorek, I.: From global governance to nationalism: the future of AI. *GIGA Focus* (2025)
128. Data Science Dojo Staff: AI governance checklist for CTOs, CIOs, and AI teams: a complete blueprint for 2025. *Data Science*

- Dojo (2025). <https://datasciencedojo.com/blog/ai-governance-checklistfor-2025/>
129. World Economic Forum: Global technology governance: a multi-stakeholder approach (White Paper). (2019). https://www3.weforum.org/docs/WEF_Global_Technology_Governance.pdf
 130. Yu, X., Wang, Q.: Tentative governance: an exploration of ethical governance strategies for emerging science and technology. *Cultures Sci.* (2024). <https://doi.org/10.1177/20966083251314585>
 131. van Rosmalen, G.: Who governs platform power? Reforming governance as an EU competition law remedy to democratic harm. *Eur. Competition J.* (2026). <https://doi.org/10.1080/17441056.2026.2675841>
 132. Dwi, M.: Ethical and psychological implications of generative AI in digital afterlife technologies: a systematic literature review on responsible inclusive innovation. *Technol. Soc.* (2025). <https://www.sciencedirect.com/science/article/pii/S2666659625000320>
 133. Benz, C., Scott-Jeffs, W., McKercher, K.A., Welsh, M., Norman, R., Hendrie, D., Locantro, M., Robinson, S.: Community-based participatory-research through co-design: supporting collaboration from all sides of disability. *Res. Involv. Engagem.* **10**(1), 47 (2024). <https://doi.org/10.1186/s40900-024-00573-3> PMID: 38730283; PMCID: PMC11084036
 134. O'Donnell, M., Collier, M., Pineda-Pinto, M., Cooper, C., Nulty, F., Castañeda, N.R.: Redefining co-design for social-ecological research and practice: a systematic literature review. *Environ. Sci. Policy.* **164**, 103998 (2025)
 135. Bujar, M., McAuslane, N., Connelly, P., Walker, S.R.: Quality decision-making practices in pharmaceutical companies and regulatory authorities: current and proposed approaches to its documentation. *Ther. Innov. Regul. Sci.* **54**(6), 1404–1415 (2020). <https://doi.org/10.1007/s43441-020-00167-7>

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